Probability Theory and Statistics

Exercise 8 solution

10.27. – 10.31.

Covariance, Joint Density Function

1. Since the two random variables are independent, their joint distribution function is the product of the two marginal distribution functions. Using the general formula for the CDF of a uniform distribution:
$F_X(x) = \dfrac{x-a}{b-a} = \dfrac{x-0}{3-0} = \dfrac{x}{3}$ if $0 \le x \le 3$, 0 if $x < 0$, and 1 if $x > 3$.
$F_Y(y) = \dfrac{y-(-1)}{4-(-1)} = \dfrac{y+1}{5}$ if $-1 \le y \le 4$, 0 if $y < -1$, and 1 if $y > 4$.
Hence the joint distribution function is

$$F_{X,Y}(x,y) = \begin{cases} 0, & \text{if } x < 0 \text{ or } y < -1, \\ \dfrac{x}{3} \cdot \dfrac{y+1}{5}, & \text{if } 0 \le x \le 3 \text{ and } -1 \le y \le 4, \\ \dfrac{x}{3}, & \text{if } 0 \le x \le 3 \text{ and } y > 4, \\ \dfrac{y+1}{5}, & \text{if } x > 3 \text{ and } -1 \le y \le 4, \\ 1, & \text{if } x > 3 \text{ and } y > 4. \end{cases}$$

The CDF takes values in $[0,1]$, so we look for level sets for $0 \le c \le 1$. For $c = 0$ and $c = 1$, the level sets follow from the above: respectively $\mathbb{R}^2 \setminus \big((0,\infty) \times (-1,\infty)\big)$ and $(3,\infty) \times (4,\infty)$. If $0 < c < 1$, then the part of the level curve $F_{X,Y}(x,y) = c$ lying in the rectangle $(0,3) \times (-1,4)$ consists of those (x,y) with $\dfrac{x(y+1)}{15} = c$, i.e. $y = \dfrac{15c}{x} - 1$, a hyperbola restricted to the rectangle. This level curve also includes two half-lines in the regions $(0,3) \times (4,\infty)$ and $(3,\infty) \times (-1,4)$, obtained as follows: in the first case $F_{X,Y}(x,y) = \dfrac{x}{3}$, so $\dfrac{x}{3} = c$, i.e. $x = 3c$; in the second case $F_{X,Y}(x,y) = \dfrac{y+1}{5}$, so $\dfrac{y+1}{5} = c$, i.e. $y = 5c - 1$.
a) Since both variables are uniformly distributed, taking a value corresponds to choosing a point uniformly at random from the rectangle $(0,3) \times (-1,4)$. The event $\{X < Y\}$ corresponds to the subset above the line $x = y$ in this rectangle. By geometric probability, the probability is the ratio of the favorable area to the total area:

$$\mathbb{P}(X < Y) = \frac{\frac{3 \cdot 3}{2} + 1 \cdot 3}{3 \cdot 5} = \frac{7.5}{15} = \frac{1}{2}.$$

b) The event $\{X + Y = 1\}$ corresponds to the line $x + y = 1$, i.e. $y = 1 - x$. A straight line has area 0, hence the probability is 0.
c) Again the probability is the ratio of the favorable area to the area of the whole rectangle. The favorable region is the part of $(0,3) \times (-1,4)$ lying below the hyperbola $xy = 1$, i.e. $y = \dfrac{1}{x}$. This splits into three parts: the rectangle $(0,3) \times (-1,0)$, the rectangle $(0,\frac{1}{4}) \times (0,4)$,

and the remaining part of $(\frac{1}{4}, 3) \times (0, 4)$ under the hyperbola, whose area we obtain by integration:

$$\mathbb{P}(XY < 1) = \frac{3 \cdot 1 + \frac{1}{4} \cdot 4 + \int_{\frac{1}{4}}^{3} \frac{1}{x}\, dx}{3 \cdot 5} = \frac{3 + 1 + [\ln x]_{\frac{1}{4}}^{3}}{15} = \frac{4 + \ln 3 - \ln \frac{1}{4}}{15} = \frac{4 + \ln 12}{15} \approx 0.4323.$$

2. The event $\{V < Z\}$ in terms of the original variables (substituting $Z = 2X + 1$ and $V = 3Y$) is
$$\{V < Z\} \equiv \{3Y < 2X + 1\} \equiv \left\{Y < \frac{2}{3}X + \frac{1}{3}\right\}.$$
Since X and Y are independent and uniform on $(0, 1)$, choosing their values corresponds to selecting a point uniformly in the unit square. Thus we have a geometric probability problem: the probability equals the area of the favorable set divided by the area of the unit square.
The favorable set is the region beneath the line $y = \frac{2}{3}x + \frac{1}{3}$ within the unit square, a trapezoid of area $\left(\frac{1}{3} + 1\right) \cdot 1 \cdot \frac{1}{2} = \frac{2}{3}$.
This equals the probability (the unit square has area 1). Hence $\mathbb{P}(V < Z) = \dfrac{2}{3}$.
3. **a)** $\mathbb{P}(X + Y < 1) = \mathbb{P}(Y < 1 - X)$ is obtained by integrating the joint density over the set $\{(x, y) \in (0, 1)^2 : y < 1 - x\}$, i.e. integrate with $x = 0 \ldots 1$ and $y = 0 \ldots 1 - x$ (or in the reverse order: $y = 0 \ldots 1$, $x = 1 - y \ldots 1$). Thus
$$\begin{aligned}\mathbb{P}(X + Y < 1) &= \int_0^1 \int_0^{1-x} f_{X,Y}(x, y)\, \mathrm{d}y\mathrm{d}x = \int_0^1 \int_0^{1-x} 2(x^3 + y^3)\, \mathrm{d}y\mathrm{d}x \\ &= \int_0^1 \left[2\frac{y^4}{4} + 2x^3 y\right]_{y=0}^{1-x} \mathrm{d}x = \int_0^1 \left(\frac{2(1-x)^4}{4} + 2x^3 - 2x^4\right) \mathrm{d}x \\ &= \left[\frac{1}{2}\frac{(1-x)^5}{-5} + \frac{2x^4}{4} - \frac{2x^5}{5}\right]_0^1 = 0 + \frac{1}{2} - \frac{2}{5} + \frac{1}{10} = \frac{1}{5}.\end{aligned}$$
(Quick note for those less familiar with multivariable calculus: integrate inside–out, treating the other variable as constant in the inner integral, and always indicate which variable you integrate with respect to.)
b) Since $Y > 0$ almost surely,
$$\begin{aligned}\mathbb{P}(X^2 < Y) = \mathbb{P}(X < \sqrt{Y}) &= \int_0^1 \int_0^{\sqrt{y}} f_{X,Y}(x, y)\, \mathrm{d}x\mathrm{d}y = \int_0^1 \int_0^{\sqrt{y}} 2(x^3 + y^3)\, \mathrm{d}x\mathrm{d}y \\ &= \int_0^1 \left[\frac{2x^4}{4} + 2y^3 x\right]_0^{\sqrt{y}} \mathrm{d}y = \int_0^1 \left(\frac{2y^2}{4} + 2y^{7/2}\right) \mathrm{d}y \\ &= \left[\frac{y^3}{6} + \frac{4}{9}y^{9/2}\right]_0^1 = \frac{1}{6} + \frac{4}{9} = \frac{11}{18}.\end{aligned}$$
c) The marginal density is obtained by integrating out the other variable from $-\infty$ to ∞ (practically, over the support). Since $f_{X,Y}(x, y) = 0$ whenever $x \notin (0, 1)$, we have $f_X(x) = 0$ if $x \leq 0$ or $x \geq 1$. For $0 < x < 1$,
$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, \mathrm{d}y = \int_0^1 2(x^3 + y^3)\, \mathrm{d}y = \left[2x^3 y + \tfrac{2}{4}\, y^4\right]_0^1 = 2x^3 + \frac{1}{2}.$$

In summary,

$$f_X(x) = \begin{cases} 2x^3 + \dfrac{1}{2}, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

d) By symmetry,

$$f_Y(y) = \begin{cases} 2y^3 + \dfrac{1}{2}, & 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(Here we integrate with respect to x from $-\infty$ to ∞; the computation mirrors that for f_X with $x \leftrightarrow y$.)

e) $\mathbb{E}(X) = \int_{-\infty}^{\infty} x f_X(x)\,\mathrm{d}x = \int_0^1 \left(2x^4 + \frac{1}{2}x\right)\mathrm{d}x = \left[2\frac{x^5}{5} + \frac{1}{4}x^2\right]_0^1 = \frac{2}{5} + \frac{1}{4} = \frac{13}{20}$. (Equivalently, $\mathbb{E}(X) = \iint x\, f_{X,Y}(x,y)\,\mathrm{d}x\mathrm{d}y$ yields the same result.)

f) Since (X,Y) is jointly continuous, X and Y are independent iff $f_{X,Y}(x,y) = f_X(x)f_Y(y)$ for all $x, y \in \mathbb{R}$. They are *not* independent, e.g. for $x = y = \frac{1}{2}$,

$$f_X(x)f_Y(y) = \left(2x^3 + \tfrac{1}{2}\right)\left(2y^3 + \tfrac{1}{2}\right) \neq 2(x^3 + y^3) = f_{X,Y}(x,y).$$

Equivalently, $f_{Y|X}(y|x) \neq f_Y(y)$ for some (x,y) (e.g. $x = y = 1/2$).

4. The desired probability is the integral of the joint density over the favorable region. Differentiate the joint CDF with respect to both x and y to obtain the joint density:

$$f_{X,Y}(x,y) = \frac{\partial^2 F_{X,Y}(x,y)}{\partial x \partial y} = \frac{\partial^2 \frac{xy^3+x}{2}}{\partial x \partial y} = \frac{\partial \frac{y^3+1}{2}}{\partial y} = \frac{3y^2}{2}.$$

The triangle with vertices $A(0,0)$, $B(\frac{1}{2},0)$, and $C(\frac{1}{2},-\frac{1}{4})$ is bounded by the x–axis, the line $x = \frac{1}{2}$, and the line $y = -\frac{1}{2}x$, i.e. $0 < x < \frac{1}{2}$ and $-\frac{1}{2}x < y < 0$. Thus the probability that (X,Y) falls inside the triangle is

$$\int_0^{\frac{1}{2}} \int_{-\frac{1}{2}x}^{0} \frac{3y^2}{2}\,\mathrm{d}y\mathrm{d}x = \int_0^{\frac{1}{2}} \left[\frac{y^3}{2}\right]_{-\frac{1}{2}x}^{0} \mathrm{d}x = \int_0^{\frac{1}{2}} \frac{\frac{1}{8}x^3}{2}\,\mathrm{d}x = \left[\frac{x^4}{64}\right]_0^{\frac{1}{2}} = \frac{1}{1024}.$$

(Note: since the limits for y depend on x, integrate first with respect to y, then with respect to x.)

5. Independence holds iff $f_{X,Y}(x,y) = f_X(x)f_Y(y)$ for all $x, y \in \mathbb{R}$, since X and Y are jointly continuous. It suffices to consider $x, y \in (0,1)$, as elsewhere the joint density is 0 (and at least one marginal is 0 as well).

Compute the marginals by integrating out the other variable:

$$\begin{aligned} f_X(x) &= \int_0^1 f_{X,Y}(x,y)\,\mathrm{d}y = \int_0^1 \left(a(4x+y) + bxy + \tfrac{2}{5}\right)\mathrm{d}y \\ &= \left[4axy + \tfrac{ay^2}{2} + bx\tfrac{y^2}{2} + \tfrac{2}{5}y\right]_0^1 = 4ax + \frac{a}{2} + \frac{bx}{2} + \frac{2}{5}, \end{aligned}$$

and

$$f_Y(y) = \int_0^1 f_{X,Y}(x,y)\,\mathrm{d}x = \int_0^1 \left(a(4x+y) + bxy + \tfrac{2}{5}\right)\mathrm{d}x$$
$$= \left[2ax^2 + axy + bx^2y + \tfrac{2}{5}y\right]_0^1 = 2a + ay + \frac{by}{2} + \frac{2}{5}.$$

Thus independence holds iff

$$a(4x+y) + bxy + \frac{2}{5} = \left(4ax + \tfrac{a}{2} + \tfrac{bx}{2} + \tfrac{2}{5}\right)\left(2a + ay + \tfrac{by}{2} + \tfrac{2}{5}\right).$$

Comparing constant terms on both sides gives

$$\frac{2}{5} = a^2 + \frac{1}{5}a + \frac{4}{5}a + \frac{4}{25} \quad \Rightarrow \quad a^2 + a - \frac{6}{25} = 0.$$

The solutions are

$$a_{1,2} = \frac{-1 \pm \sqrt{1 + \frac{24}{25}}}{2} = \frac{-1 \pm \frac{7}{5}}{2},$$

so $a_1 = 1/5$ and $a_2 = -6/5$. The latter is invalid since $f_X(x)$ must be nonnegative for all $x \in (0,1)$, even near 0, but

$$\lim_{x \downarrow 0}\left(4ax + \frac{a}{2} + \frac{bx}{2} + \frac{2}{5}\right) = \frac{a}{2} + \frac{2}{5} = -\frac{3}{5} + \frac{2}{5} < 0 \quad \text{if } a = -\frac{6}{5}.$$

Hence $a = 1/5$. To find b, compare the xy-terms on both sides (substitute $a = 1/5$):

$$b = \frac{b^2}{4} + \frac{4}{25} + \frac{b}{10} + \frac{2}{5}b \quad \Rightarrow \quad \frac{b^2}{4} + \frac{4}{25} - \frac{1}{2}b = 0.$$

Solving,

$$b_{1,2} = \frac{\frac{1}{2} \pm \sqrt{\frac{1}{4} - \frac{4}{25}}}{1/2} = 2\left(\frac{1}{2} \pm \frac{3}{10}\right),$$

so $b_1 = \dfrac{8}{5}$ and $b_2 = \dfrac{2}{5}$. We must decide which one is valid. Check, e.g., whether $\int_0^1 f_X(x)\,\mathrm{d}x = 1$:

$$\int_0^1 f_X(x)\,\mathrm{d}x = \left[\frac{4ax^2}{2} + \frac{ax}{2} + \frac{bx^2}{4} + \frac{2x}{5}\right]_0^1 = 2a + \frac{a}{2} + \frac{b}{4} + \frac{2}{5} = \frac{2}{5} + \frac{1}{10} + \frac{1}{10} + \frac{2}{5} = 1,$$

when $b = 2/5$. The integral is not 1 for $b = 8/5$. Therefore the correct values are

$$a = \frac{1}{5}, \qquad b = \frac{2}{5}.$$

(Checking that $f_{X,Y}(x,y) = f_X(x) f_Y(y)$ holds for all $x, y \in (0,1)$ is left to the reader; compare also the coefficients of x and y.)

6. Since (X,Y) is uniform over the given region Ω, the joint density is a constant $c > 0$ on Ω and 0 elsewhere, with c chosen so that $\iint_{\mathbb{R}^2} f_{X,Y}(x,y)\,\mathrm{d}x\mathrm{d}y = 1$, i.e. $c = 1/T_\Omega$.
Solution 1: The region is $\{(x,y) : 0 < x < \pi,\ 0 < y < \sin x\}$, hence

$$T_\Omega = \int_0^\pi \int_0^{\sin(x)} 1\,\mathrm{d}y\mathrm{d}x = \int_0^\pi \sin(x)\,\mathrm{d}x = [-\cos(x)]_0^\pi = -(-1) - (-1) = 2,$$

so $c = 1/2$.

Solution 2: The area is the area under $\sin x$ from $x = 0$ to $x = \pi$: $T_\Omega = \int_0^\pi \sin(x)\,\mathrm{d}x = [-\cos(x)]_0^\pi = 2$, hence $c = 1/2$.

Therefore,

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,\mathrm{d}y = \int_0^{\sin(x)} \frac{1}{2}\,\mathrm{d}y = \frac{1}{2}\sin(x), \quad 0 < x < \pi,$$

and $f_X(x) = 0$ otherwise.

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7. **a)** The pair (X, Y) always consists of two distinct integers from $\{1, 2, 3\}$, all equally likely (e.g., the sample space can be modeled as permutations of size 2 without replacement). Thus $\mathbb{P}(X = k, Y = l) = \dfrac{1}{3 \cdot 2} = \dfrac{1}{6}$ for $k \neq l$.

$$\mathbb{E}(XY) = 1{\cdot}2{\cdot}\mathbb{P}(X=1,Y=2) + 1{\cdot}3{\cdot}\mathbb{P}(X=1,Y=3) + 2{\cdot}3{\cdot}\mathbb{P}(X=2,Y=3) + 2{\cdot}1{\cdot}\mathbb{P}(X=2,Y=1) + 3{\cdot}1{\cdot}\mathbb{P}$$

$\mathbb{E}(X) = \dfrac{1}{3}(1 + 2 + 3) = 2$, since X is uniformly distributed over $\{1, 2, 3\}$ (this also follows from the joint distribution). Similarly $\mathbb{E}(Y) = 2$ (same marginal distribution by symmetry or direct calculation, e.g. $\mathbb{P}(Y = 1) = \mathbb{P}(Y = 1, X = 2) + \mathbb{P}(Y = 1, X = 3) = \frac{1}{3}$). Hence

$$\operatorname{cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y) = \frac{11}{3} - 4 = -\frac{1}{3}.$$

b) $\operatorname{cov}(X, X) = \mathbb{D}^2(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2$. Since

$$\mathbb{E}(X^2) = 1^2 \cdot \mathbb{P}(X = 1) + 2^2 \cdot \mathbb{P}(X = 2) + 3^2 \cdot \mathbb{P}(X = 3) = \frac{14}{3},$$

we get $\mathbb{D}^2(X) = \dfrac{14}{3} - 4 = \dfrac{2}{3}$.

c) By identical marginals, $\operatorname{cov}(Y, Y) = \mathbb{D}^2(Y) = \mathbb{D}^2(X) = \dfrac{2}{3}$.

d) Since their covariance is nonzero, X and Y are not independent. (One can also argue directly from the joint pmf.)

8. Using $\operatorname{cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$ and bilinearity,

$$\operatorname{cov}(X, X^2 - 4X + 4) = \operatorname{cov}(X, X^2 - 4X) = \operatorname{cov}(X, X^2) - \operatorname{cov}(X, 4X) = \mathbb{E}(X^3) - \mathbb{E}(X)\mathbb{E}(X^2) - 4\,\mathbb{D}^2(X).$$

With the given moments,

$$14 - 2 \cdot 5 - 4\,(5 - 4) = 14 - 10 - 4 = 0.$$

This is correct provided the covariance exists, which requires both $\mathbb{E}(X^2) < \infty$ and $\mathbb{E}\Big((X^2 - 4X)^2\Big) < \infty$. We know $\mathbb{E}(X^2)$ is finite; $\mathbb{E}\Big((X^2 - 4X)^2\Big) = \mathbb{E}(X^4) - 8\mathbb{E}(X^3) + 16\mathbb{E}(X^2)$ may fail to exist unless $\mathbb{E}(X^4) < \infty$. If $\mathbb{E}(X^4) < \infty$, the answer is 0; otherwise the covariance does not exist.

If $(X-2)^2$ is almost surely constant, it is independent of any random variable, in particular of $X-2$. We show that if $(X-2)^2$ is *not* a.s. constant, then $X-2$ and $(X-2)^2$ are *not* independent. Assume (for contradiction) they are independent. Then by Lemma 10.1.5, for any continuous real g, h, $g(X-2)$ and $h\big((X-2)^2\big)$ are independent. Let $g(x) = |x|$ and $h(x) = \sqrt{x}$. Then $g(X-2) = h\big((X-2)^2\big) = |X-2|$, so $|X-2|$ is independent of itself. By the definition of independence, for any real t the events $\{|X-2| < t\}$ and itself are independent:

$$\mathbb{P}(|X-2| < t) = \mathbb{P}(\{|X-2| < t\} \cap \{|X-2| < t\}) = \mathbb{P}(|X-2| < t)\,\mathbb{P}(|X-2| < t) = \mathbb{P}(|X-2| < t)^2.$$

Thus $F_{|X-2|}(t) = \mathbb{P}(|X-2| < t)$ equals its square, hence is 0 or 1 for all t, i.e. $|X-2|$ is a.s. constant, so $(X-2)^2$ is also a.s. constant—a contradiction. Therefore $X-2$ and $(X-2)^2$ are not independent unless $(X-2)^2$ is a.s. constant.

Remark: The same argument works with $g(x) = x^2$ and $h(x) = x$. *Remark 2:* Since $\operatorname{cov}(X, X^2 - 4X) = \operatorname{cov}(X-2, (X-2)^2)$, this gives another example of zero covariance without independence (provided there exists an X with the given first three moments).

9. Again use $\operatorname{cov}(X,Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$. Note that if $X = 2$, then $Y = 1$, so $XY = 2$. On the event $\{X = 1\}$, the outcomes are $(1,2), (1,3), \ldots, (1,6)$ and $(2,1), (3,1), \ldots, (6,1)$, ten cases in the classical 36-point sample space. For $(2,1), \ldots, (6,1)$ we have $Y = 1$; for $(1,2), \ldots, (1,6)$, Y is the (non–one) second die outcome. Hence

$$\begin{aligned}\mathbb{E}(XY) &= 2 \cdot 1 \cdot \mathbb{P}(X = 2, Y = 1) + 1 \cdot 1 \cdot \mathbb{P}(\{(2,1), \ldots, (6,1)\}) \\ &\quad + 1 \cdot 2 \cdot \mathbb{P}(\{(1,2)\}) + \cdots + 1 \cdot 6 \cdot \mathbb{P}(\{(1,6)\}) \\ &= \frac{2}{36} + \frac{5}{36} + \frac{2+3+4+5+6}{36} = \frac{27}{36}.\end{aligned}$$

Also $\mathbb{P}(X = 2) = \frac{1}{36}$, $\mathbb{P}(X = 1) = \frac{10}{36}$, and $\mathbb{P}(X = 0) = \frac{25}{36}$, so $\mathbb{E}(X) = \frac{10}{36} \cdot 1 + \frac{1}{36} \cdot 2 = \frac{1}{3}$ (alternatively, by linearity of expectation over two dice, the expected number of ones is $2 \cdot \frac{1}{6}$). Further, $\mathbb{E}(Y) = \frac{7}{2}$ (the well-known mean of a fair die). Thus

$$\operatorname{cov}(X,Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y) = \frac{27}{36} - \frac{7}{2} \cdot \frac{1}{3} = \frac{14}{18} - \frac{27}{36} = -\frac{15}{36} \approx -0.4167.$$

10. Using the alternative covariance identity and linearity of expectation,

$$\begin{aligned}\operatorname{cov}(X+Y, X-Y) &= \mathbb{E}\Big((X+Y)(X-Y)\Big) - \mathbb{E}(X+Y)\mathbb{E}(X-Y) \\ &= \mathbb{E}(X^2 - Y^2) - (\mathbb{E}(X) + \mathbb{E}(Y))(\mathbb{E}(X) - \mathbb{E}(Y)) \\ &= \mathbb{E}(X^2) - \mathbb{E}(Y^2) - \mathbb{E}(X)^2 - \mathbb{E}(Y)\mathbb{E}(X) + \mathbb{E}(Y)\mathbb{E}(X) + \mathbb{E}(Y)^2 \\ &= \mathbb{D}^2(X) - \mathbb{D}^2(Y) = 0,\end{aligned}$$

since X and Y have the same (finite) variance.